

Are There Ideological Asymmetries in Intergroup Bias? A Minimal Groups Approach

Benjamin C. Ruisch¹, Rajen A. Anderson², Youssef Amin,³ Angelika Stefan⁴, Amy R. Krosch^{5*}

¹ School of Psychology, University of Kent, UK

² Marketing Department, University of Leeds, UK

³ Department of Psychology, Princeton University, U.S.A.

⁴ Bundeswehr University, Munich, Germany

⁵ Department of Psychology, Cornell University, U.S.A.

* Corresponding Author, amy.krosch@cornell.edu

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Abstract

The divide between political liberals and conservatives is rapidly growing. Several influential theories contend that this divide hinges on orientations towards social groups, such that conservatives (versus liberals) show a greater tendency to favor their “ingroups” and discriminate against “outgroups.” However, other theories contend that liberals and conservatives do not differ in their degree of intergroup bias. Both perspectives have received empirical support, and the debate has reached a standstill. We argue that this theoretical and empirical stalemate stems from inherent limitations of examining attitudes towards real-world social groups—a strategy used by both sides of the debate. Drawing on social identity theory, we propose an alternative approach—using “minimal groups” (i.e., experimentally constructed groups)—to determine whether and why ideological differences in intergroup bias may exist. Finding support for either account will help resolve this longstanding debate and deepen our understanding of the psychological underpinnings of political ideology.

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Introduction

Tensions between the political right and left have grown in recent years, with greater polarization, increased animosity, and less willingness to engage with those of the opposing ideology or political party^{1–5}. Indeed, in many nations, the political divide is now among the most contentious divisions of modern society^{3,4,6,7}. But what underlies these conflicting worldviews, and what factors influence whether a person adopts a more liberal versus conservative ideology?

Several leading theories suggest that the left-right divide hinges on orientations towards social groups^{8–10}. More politically conservative people are argued to possess a suite of basic psychological traits and motivations that lead them to exhibit greater “intergroup bias”—a relatively stronger preference for their social “ingroups” (i.e., the groups to which they belong) and relatively stronger prejudice/animosity towards social “outgroups” (i.e., groups to which they do not belong). However, the empirical evidence for this highly influential theoretical perspective has been mixed. While some research has found that conservatives/right-wing individuals generally exhibit greater intergroup bias^{11–14}, other research has suggested there are no ideological differences in the overall degree or extent of intergroup bias^{15–17}. Despite decades of research, this debate appears little closer to being resolved.

In the present work, we draw on theory and methodology from social cognition to propose an alternative approach to understanding whether conservatives are more predisposed than liberals to favor their own social groups over others. This approach is centered on the “minimal groups” paradigm^{18,19}, which was designed to strip away the complex social information associated with real-world social groups in order to examine the most basic forms of intergroup bias^{20,21}. Through pilot data and a series of five studies, we aim to provide a more decisive test of ideological differences in intergroup bias, utilizing several distinct indices of intergroup favoritism, including more diverse participant samples, and systematically manipulating features of the intergroup context to understand the conditions under which ideological differences in intergroup bias are most likely to emerge.

The “asymmetrical intergroup bias hypothesis”—that more conservative belief systems are associated with greater intergroup bias—was first proposed in the early 1950s. Adorno and colleagues²² argued that individuals with a more “authoritarian” (i.e., conservative¹¹) worldview were generally predisposed to hold greater prejudice against social outgroups. Support for this argument rested on two central empirical observations: 1) prejudices are positively correlated, such that individuals who are more prejudiced against one minority group tend to be more prejudiced against other minority groups, and 2) an authoritarian belief system strongly predicts this form of “generalized intergroup bias.” This perspective has proven remarkably influential in political psychology in the intervening decades, and the

idea that conservatives/those on the political right are fundamentally more prejudiced against social outgroups is central to many theories of political ideology and related belief systems.^{11,12,23–25}

Other programs of research independently conclude that intergroup attitudes are central to the liberal-conservative ideological divide. For example, moral foundations research suggests ideological differences in the relative weight assigned to ingroup-protective moral values⁸, such that political conservatives more strongly endorse moral values theorized to maintain ingroup cohesion and cooperation (e.g., ingroup loyalty and respect for authority), whereas liberals tend to hold a more individualistic view of morality that does not privilege ingroups over outgroups^{26,27}. Similarly, evolutionary psychologists have argued that conservatism is a constellation of evolved psychological mechanisms designed to facilitate coalition building, maintain ingroup cohesion, and ward off threats by outgroups. According to these accounts, conservatives are “in-group specialists”¹⁰ who possess a set of psychological traits that predispose them to form tighter-knit social groups, to exhibit greater favoritism toward their own social ingroups, and to be more likely to derogate and aggress against outgroups^{27–31}. Conversely, liberals are argued to be “outgroup specialists”¹⁰ who tend to make less of a psychological distinction between “us” and “them” and display relatively less favoritism towards their own groups—an orientation that may allow for greater intergroup cooperation and exposure to new ideas and practices.

Importantly, these theoretical perspectives predict that conservatives’ greater ingroup favoritism should extend to social ingroups and outgroups writ large—regardless of the origin or nature of those groups. However, empirical support for the asymmetrical intergroup bias hypothesis has so far come exclusively from studies examining attitudes towards real-world social groups (particularly racial/ethnic, religious, and cultural groups), and the full extent of conservatives’ ostensibly greater intergroup bias therefore remains an open question.

In recent years, a rapidly growing body of research has challenged the asymmetrical intergroup bias hypothesis and has raised doubt about whether conservatives truly exhibit greater intergroup bias in general. This work supports a “symmetrical intergroup bias hypothesis” and instead suggests that the relationship between ideology and intergroup bias is limited to *specific* kinds of social groups.

Early support for this perspective came from the finding that social dominance orientation and right-wing authoritarianism (SDO and RWA), two broad belief systems closely associated with political conservatism^{32–34}, do not predict greater negativity toward all minority groups^{13,32,35,36}. Rather, these belief systems only predict greater prejudice towards minority groups that are perceived to be threatening or low in status^{32,35,36}. Further evidence against a general association between conservatism and intergroup bias comes from recent research and theory that has raised doubts about the existence

of ideological differences in moral foundation endorsement. In particular, Schein and colleagues argue that the apparent association between conservatism and ingroup-centric morality is simply an artifact of the specific social groups that have been examined in past research^{37–39}. They contend that researchers have selectively sampled social groups that conservatives tend to value more, and in doing so has “conflated conservatism and [ingroup-centric] morality from the very beginning”³⁹. Further, they argue that there are social groups for which liberals, too, value loyalty and respect for authority, and that expanding the field of investigation to include these groups would attenuate or even eliminate ideological differences in ingroup-protective moral beliefs^{39,40}.

Perhaps the strongest evidence against a general association between conservatism and intergroup bias comes from a rapidly growing body of research on the “ideological conflict hypothesis”^{15–17,41–43}. Drawing from earlier work on liking and dissimilarity^{44–46}, this perspective contends that all individuals—conservative or liberal—tend to dislike others who hold beliefs and values that are dissimilar from their own. This research suggests that the apparent association between conservatism and outgroup prejudice is driven by the fact that past research has focused almost exclusively on attitudes towards low-status minority groups—groups that tend to themselves be (perceived as) liberal or to be emblematic of liberal ideals and political causes (e.g., African Americans, gay men, Muslims). Because conservatives’ beliefs and values conflict with the (real or perceived) beliefs and values of these groups, conservatives tend to express greater prejudice towards these groups. This “worldview conflict,” the authors argue, explains why conservatives have appeared to be more prejudiced in past research. Supporting this argument, they show that when the range of target groups includes non-liberal social groups (i.e., groups perceived to be politically conservative in their beliefs and/or emblematic of right-wing ideals and causes, like the elderly or fundamentalist Christians), the prejudice gap between liberals and conservatives disappears, and “liberals’ and conservatives’ prejudices manifest to roughly equal degrees”¹⁷.

Together, this body of research supporting the “symmetrical intergroup bias hypothesis” provides important insight into the nature and extent of ideological differences in intergroup cognition. By using a more comprehensive array of social groups, this research convincingly demonstrates that the liberal-conservative prejudice gap is likely overstated, the psychological processes underlying intergroup discrimination are often symmetrical, and liberals and conservatives, under some circumstances, can show similar levels of prejudice. However, this work, too, has relied exclusively on the examination of attitudes toward real-world social groups, and therefore this research – like the research supporting the asymmetrical intergroup bias hypothesis – is vulnerable to various confounding factors stemming from

the complex nature of these target groups. As a result, neither branch of work can decisively answer the question that lies at the heart of this debate: Are there ideological differences in intergroup bias *in general*?

The primary issue with examining real-world groups is that liberals and conservatives differ in the kinds of social ingroups and outgroups to which they belong⁴⁷. For example, political conservatives are more likely to belong to high-status, high-power, “majority” groups (e.g., racial/ethnic, religious, sexual orientation, and cultural groups⁴⁸), while liberals are more likely to belong to (a range of different) minority groups. Aside from their obvious numerical, status, and power differences, these groups also differ greatly in their history, culture, and salience (e.g., Whiteness, heterosexuality, and maleness are often “invisible”^{49,50}), as well as the social norms surrounding expressing (dis)favoritism towards them^{51,52}. Further, liberals and conservatives operate in largely independent social spheres⁴⁷. This means that not only the specific social groups—but even the domain of intergroup relations—will also differ substantially between those on the left and right. For example, to draw on (largely empirically supported) stereotypes^{25,47}, how are we to compare conservatives’ relative preference for their fellow bird hunters, football teams, and NASCAR enthusiasts to liberals’ affinity for their fellow latte drinkers, eco-tourists, and contemporary art aficionados? These groups are incommensurate not only in their origins, history, and purpose, but also in the deeper features that characterize the intergroup environments in which they are embedded, such as the nature of intergroup interaction (e.g., cooperative vs competitive), the criteria for membership (e.g., volitional vs. not), and numerous contextual/situational features of the interaction sphere (e.g., the presence vs. absence of intergroup threat). Thus, any examination of intergroup bias using real-world social groups will be confounded by ideological differences in group belonging, and any apparent (a)symmetries in intergroup bias could stem from these confounding factors rather than (a lack of) general ideological differences in intergroup bias.

In the proposed research, we advance this debate by drawing on theory and research from the social cognition literature to examine the most fundamental aspects of group-based cognition and intergroup discrimination. Specifically, we use variations of the “minimal groups paradigm” – designed to test the minimum necessary conditions to elicit intergroup bias^{19,20}. In this paradigm, participants are assigned to novel social groups based on arbitrary criteria (e.g., painting preferences) or even completely at random (e.g., a coin toss). Participants have no interaction with the members of either group, all individuals remain anonymous, and there is no possibility of instrumental gain for participants themselves – truly minimal conditions.

A large body of research reveals that even for these arbitrary groups, people nonetheless show a preference for their social ingroup: They exhibit greater psychological identification with their own group, evaluate their ingroup (vs. outgroup) more positively, and show intergroup bias in allocations of money and other resources^{18–21,53–55}. Research examining the psychological mechanisms underlying minimal intergroup bias suggests that simply being categorized as part of a social group creates an important psychological distinction between “us” and “them,” which in turn provides a motivation to maximize the relative standing of one’s own (vs. another) social group^{18,56,57}. This maximization of the relative status, position, or power of one’s social ingroup can be achieved through ingroup-favoring cognition or behavior, such as evaluating ingroup (vs. outgroup) members more positively (e.g., as being superior in intelligence, personality, or other traits), or allocating greater rewards or resources to ingroup (vs. outgroup) members^{18,56,58,59}.

Importantly, there is also substantial variability in the degree to which different individuals exhibit this bias^{60–64}. That is, the “us versus them” distinction seems to be more salient, central, or otherwise important for some individuals than others, with some people exhibiting pronounced favoritism for their own (vs. other) social groups, and others showing a more egalitarian intergroup orientation. The existence of individual differences in basic intergroup bias—even for these perfectly equivalent social groups—suggests that some individuals may simply exhibit greater intergroup bias in general. The form of intergroup bias that is captured in the minimal groups paradigm—reflecting a more general tendency to favor (even relatively inconsequential) ingroups over outgroups—appears to neatly parallel the kind of basic, generalized intergroup bias that has often been argued to characterize the liberal-conservative divide. Indeed, researchers from both the symmetrical intergroup bias perspective^{15,65} and the asymmetrical intergroup bias perspective^{66,67} have suggested that this paradigm may provide a means of determining whether general ideological differences in intergroup bias may exist.

Of course, even if general ideological differences in intergroup bias do exist, it is likely true that individuals who exhibit greater “minimal intergroup bias” won’t always exhibit greater bias for all social groups. Indeed, many factors shape real-world intergroup attitudes (e.g., material concerns⁶⁸, perceived threat⁶⁹, and ingroup identification⁷⁰), particularly for highly contentious, “hot-button” (e.g., racial, religious, ideological) intergroup divides. If sufficiently strong these factors may obscure or “wash out” individual differences in the more general form of intergroup bias examined with the minimal groups paradigm. However, by taking an approach that first strips away these motivations then systematically layers them on, we hope to gain a better perspective on whether, when, and why ideological differences in intergroup bias may emerge.

Despite the tremendous amount of minimal groups research (as of September 2021, a PsycINFO search for “minimal groups” returns over 1,500 results), we were unable to find any published examination of the relationship between political ideology and minimal group cognition or behavior. Moreover, the few scattered studies that have examined the relationship between minimal intergroup bias and individual difference measures related to ideology (social dominance orientation and right-wing authoritarianism) found no significant relationship.^{71,72,73} Given the strong bias toward the publication of statistically significant results⁷⁴, the existence of null effects in the literature (and no significant effects) seems to speak against the existence of ideological differences in minimal intergroup bias. However, we argue that the question of whether ideological differences exist in minimal intergroup bias is still deserving of more systematic examination. Indeed, the null results of the previous studies could reflect factors such as the absence of a direct measure of liberal-conservative political orientation, exclusive reliance on college student samples, and/or small sample sizes and low statistical power. In our pilot and preregistered studies, we therefore aim to overcome these limitations and more conclusively determine whether there are ideological differences in minimal intergroup bias, such that conservatives display greater bias than do liberals. If so, this will offer support for the asymmetrical intergroup bias hypothesis. In the absence of such an effect, we will conduct inferiority tests to determine whether practically meaningful ideological differences in minimal intergroup bias truly do not exist (or, at least, are too small to be practically significant⁷⁵)—which would provide support for the symmetrical intergroup bias perspective. (See Table 1 for a full list of hypotheses, including primary and secondary.)

In our pilot data, we provided an initial test of the asymmetrical intergroup bias hypothesis using a simple minimal groups paradigm and brief measures of ideology and intergroup bias. In our preregistered studies, we test this hypothesis using a more comprehensive measure of political ideology and three different outcome measures: ingroup identification, evaluative ratings, and allocations. To assess generalizability, we examine the nature of these effects in a sample matched to the US population on major demographic characteristics (Study 1A) as well as on Mechanical Turk (Study 1B). Unlike research using real social groups, this paradigm allows for methodologically “clean” and systematic manipulations of the intergroup environment to precisely test the conditions under which ideological differences in intergroup bias may be most likely to emerge, as well as the psychological processes associated with these effects. To this end, we systematically manipulate group relevance/importance (Study 2) and intergroup cooperation versus competition (Study 3) to explore these potentially moderating factors. We also examine multiple possible mediators (Study 2) to understand which factors may statistically account for any observed ideological differences in intergroup

bias. Finally, we use an implicit measure of ingroup identification (Study 4) to determine whether ideological differences in intergroup bias—under whatever conditions, if any, they may emerge—are relatively more automatic and uncontrollable versus more deliberative in nature. However, the participant pool of our designated data collection platform (Prolific) did not have a sufficient number of remaining participants after we completed Studies 1 through 3. Therefore, on the advice of the NHB editorial team, we did not conduct this final study.

Across these studies, our original preregistered analytic plan was to test for ideological differences in intergroup bias using standard null hypothesis significance testing, complemented by “inferiority tests” as necessary to more conclusively determine if such effects either did not exist or were too small to be practically significant⁷⁵. (See Table 1 for a full list of hypotheses, including primary and secondary.) However, after consulting with the editors and reviewers during the data collection process, we later adopted an approach instead relying on Bayesian analysis, as detailed further below. This Bayesian approach allowed us to assess the strength of the evidence that the observed data either supported or opposed the existence of a practically significant effect—that is, meaningful ideological differences in intergroup bias. In our Results section, we report both sets of analyses to ensure both interpretability and best-practices.

In our first two studies (Studies 1a and 1b), we provided a general test of ideological differences in intergroup bias under the most minimal of conditions. To do so, we used one of the most widely used and well-validated versions of the minimal groups paradigm: the “overestimator/underestimator task.” In this task, participants viewed a series of 10 images depicting randomly generated constellations of small black dots, presented on their computer screens for 300 milliseconds each. After each image, participants estimated the number of dots that appeared.

After the estimation task, participants were told that when estimating quantities, some people tend to consistently underestimate (“Underestimators”), while other people tend to consistently overestimate (“Overestimators”). They were informed that although these two groups do not differ in their overall degree of accuracy, these individual differences appear to be quite stable, and researchers suspect that they may correlate with other aspects of personality as well. After receiving this information, participants were informed of the group to which they ostensibly belonged: either Underestimators or Overestimators (group assignment was in fact random). In this way, participants were assigned to experimentally constructed social “groups” based on ostensibly meaningful—but in actuality, wholly inconsequential—characteristics.

After learning to which group they belonged, participants completed our three dependent measures in random order: (1) ratings of identification and (2) evaluative ratings of each group—i.e., to what degree they perceived members of each group negatively (lazy, selfish, incompetent) versus positively (able, intelligent, kind)—and (3) a measure of resource allocation: a matrix-based decision task in which participants chose between different combinations of points to be assigned to their ingroup versus outgroup, which were (ostensibly) later converted into monetary payments for members of each group. Using these three measures, we assessed average intergroup bias by calculating each participant's degree of favoring their own (versus the opposing) social group in identification, evaluations, and resource allocation. Finally, we assessed participants' political orientation using a series of both self-report measures (e.g., identification as liberal versus conservative) and issue-based measures (e.g., positions on issues such as welfare benefits, abortion, and limited government).

In Study 2, we built on the results of Studies 1A and 1B by manipulating the nature of the minimal groups: Groups were either randomly assigned, based on participants' choice, or based on (ostensible) similarities in political, moral, and demographic characteristics. This allowed us to assess whether the relation between ideology and intergroup bias differs across these varying types of social groups. In this study, participants first completed an assortment of measures relating to their political and moral views and demographic traits. Importantly, embedded within these measures were three theoretically-derived mediators—that is, potential psychological explanations for the relation between ideology and intergroup bias: (1) perceptions of social norms regarding the acceptability of ingroup favoritism; (2) ingroup-protective moral values (e.g., loyalty to one's social groups); and (3) intolerance of uncertainty (i.e., the motivation to hold a clear and structured worldview). This allowed us to explore whether ideological differences in any of these traits/beliefs could account for differences in intergroup bias.

After completing these measures, participants were assigned to minimal groups and completed the same measures of intergroup bias from Study 1 (identification, evaluation, resource allocation). Critically, the nature of the groups differed by random assignment: One-third of participants were assigned to their minimal groups randomly (based on a random number generator; *Random Condition*), one-third of participants were allowed to choose the group they preferred ("The Green Group" versus "The Blue Group"; *Volitional Condition*); and one-third of participants were told that group assignment was based on their responses to the earlier questionnaires concerning politics, morality, and demographics (*Attitudes Condition*). We anticipated that the relation between ideology and intergroup bias might be amplified for more meaningful groups (e.g., those based on choice or shared attitudes).

Finally, in our third study we examined the strength of the relation between ideology and intergroup bias under different intergroup conditions: cooperative, competitive, and no interaction. This allowed us to determine whether ideological differences might be heightened under certain conditions (e.g., competition), in line with theoretical perspectives arguing that ideological beliefs are largely situationally dependent, becoming more salient or “activated” under conditions of intergroup competition or threat^{72,100}. We therefore predicted that ideological differences in intergroup bias might emerge more strongly under competitive (versus cooperative) intergroup conditions.

In this study, participants completed a problem-solving task (mental rotation problems^{101,102}), ostensibly alongside other members of their ingroup and outgroup. The nature of the task—and the conditions for receiving a bonus payment—differed between participants by random assignment: Participants in the *No Interaction Condition* were informed that each person in the study would complete the problem-solving task independently and be judged solely on the basis of their own work. Participants in the *Cooperative Interaction Condition* were informed that they would work together with the other members of their team in the problem-solving task. Participants in the *Competitive Interaction Condition* were told that their group would compete against the opposing group in the problem-solving task. Following the problem-solving task, participants completed the same measures of intergroup bias (identification, evaluation, resource allocation) from our previous studies.

Results

In Studies 1a and 1b, we began by assessing our key question of interest: Whether ideological differences in intergroup bias exist. Our results provided clear evidence of ideological differences, with conservatives expressing greater favoritism for their own versus the opposing group. The magnitude of these differences was relatively small ($\beta = .12$, $p < .001$; Fig 1.) but statistically robust: our Bayes factor ($BF_{10} = 38375.74$) suggested the evidence was ... Moreover this relation did not differ across the two samples/platforms we used (stats), the nonrepresentative MTurk sample versus the census-matched Prolific sample. (On the recommendation of the editors, we therefore used the more representative Prolific sample for all subsequent studies.)

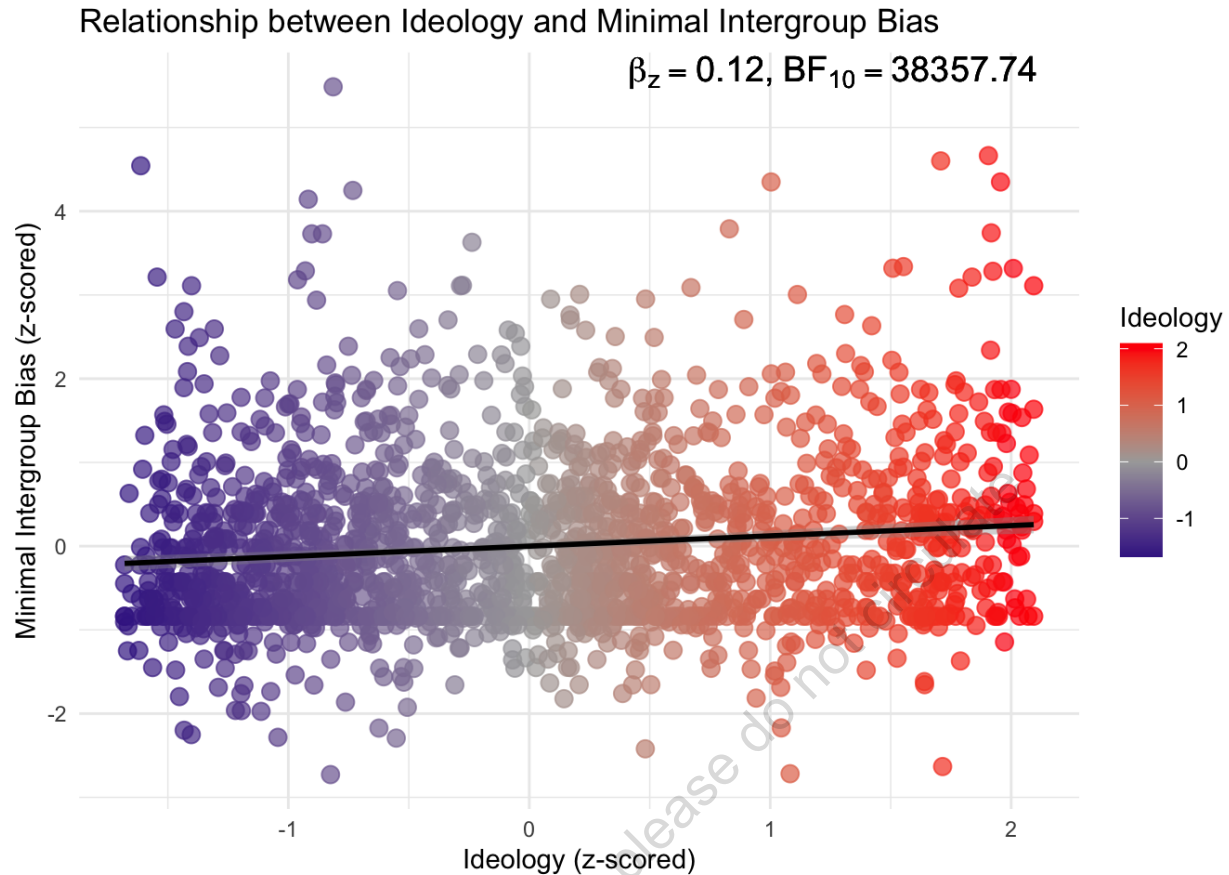


Figure 1. Scatterplot depicting the relation between political ideology and minimal intergroup bias.

In Study 2, we replicated the relation between ideology and intergroup bias, again finding that conservatives exhibited a greater preference for their ingroups over outgroups. Moreover, mediation analyses also revealed tentative evidence regarding the possible psychological mechanisms underlying this association: We found that the relation between conservatism and intergroup bias was mediated by ingroup-protective moral values (e.g., belief that loyalty is a moral value that should be upheld; indirect effect: $\beta_z = 0.197$, 95% CI [0.14, 0.25], $BF_{10} = 89825182$; 128.44% of the total effect), and, to a lesser degree, by conservatives' greater preference for clear and structured worldviews (i.e., intolerance of ambiguity; indirect effect: $\beta_z = 0.027$, 95% CI [0.013, 0.042], $BF_{10} = 1385.13$; 17.46% of the total effect).

Moreover, we found that the nature of the relation between ideology and bias did not differ as a function of whether the groups were randomly assigned or based on the participant's choice: In both cases, conservatives exhibited greater bias, with effects that were even somewhat larger than in our original studies (Random Condition: effect size and stats; Volitional Condition: effect size and stats).

Importantly, however, we found that the pattern of results differed when group assignment was ostensibly based on shared political, moral, and demographic traits: Here, the relation between political ideology and intergroup bias disappeared, with both conservatives and liberals exhibiting similar levels of bias (stats; Fig. 2).

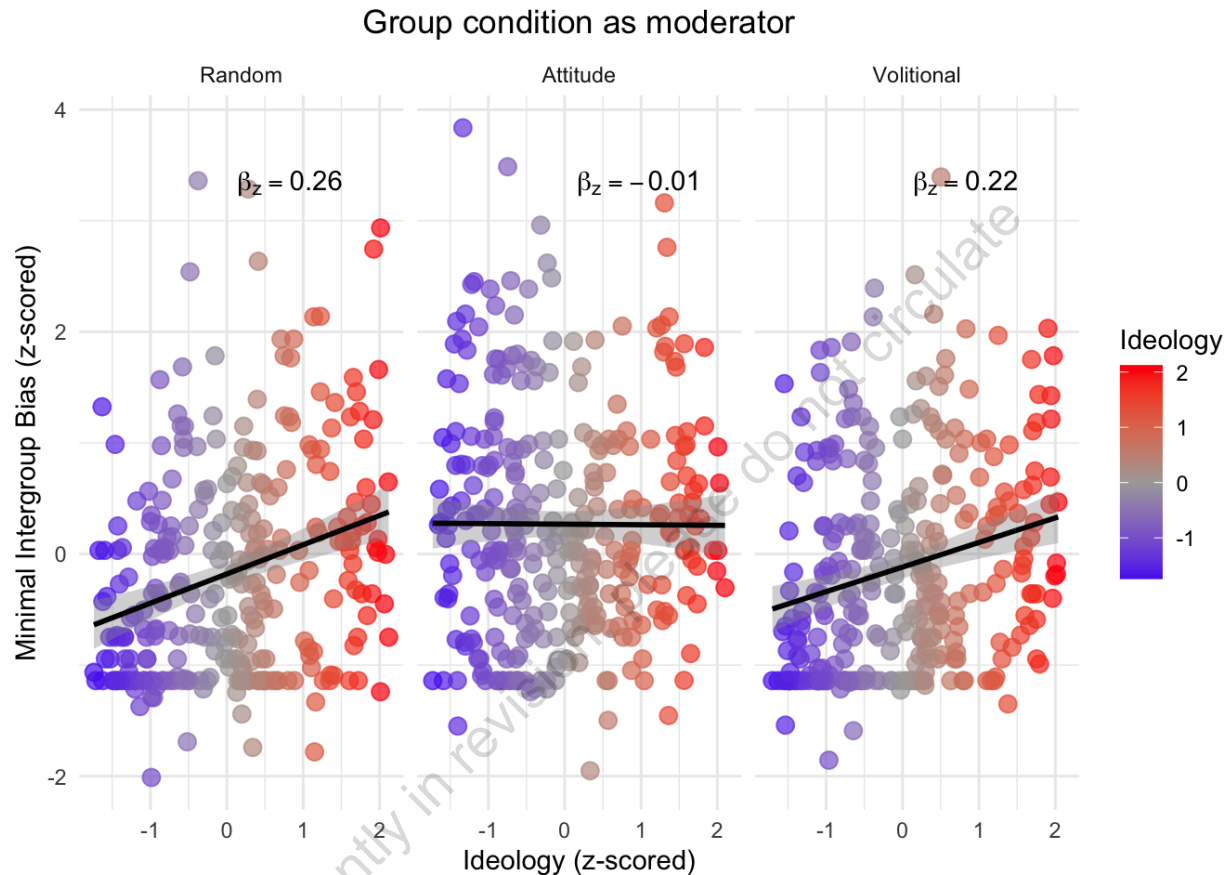


Figure 2. Scatterplot depicting the relation between ideology and intergroup bias across our three minimal group conditions: random assignment, volitional/choice-based, and attitude-based. The relation between ideology and bias emerged to a similar degree for random and volitional groups, but was eliminated for groups based on shared political and moral attitudes.

In Study 3, we again found that conservatives exhibited greater intergroup bias (stats). At odds both with leading theoretical perspectives and our own predictions, however, we found that intergroup dynamics did not affect the strength relation between political ideology and intergroup bias (stats). Rather, conservatives exhibited a similar degree of favoritism for their ingroup (relative to outgroup)

regardless of whether the problem-solving task emphasized competition between groups, cooperation within one's ingroup, or individual performance (Fig 3).

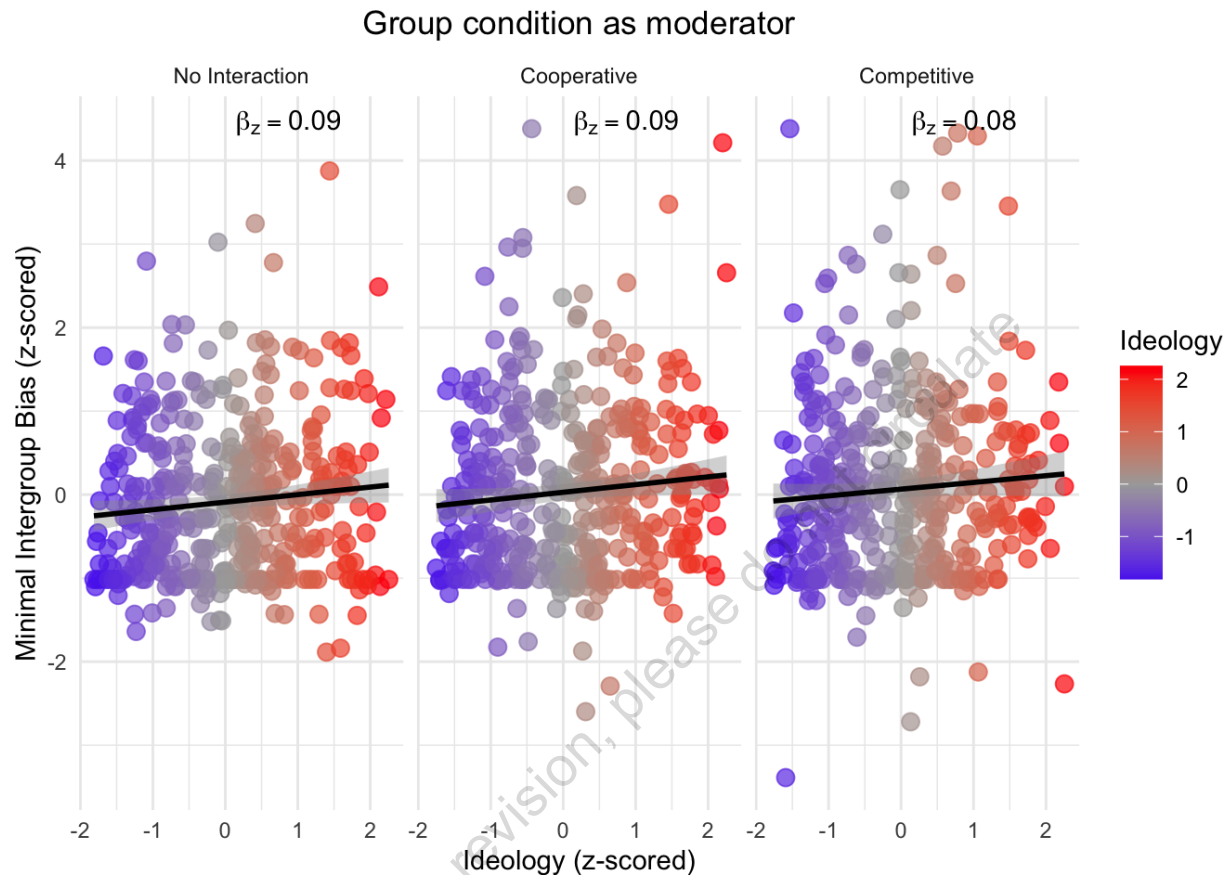


Figure 3. Scatterplots depicting the relation between ideology and intergroup bias across our three intergroup conditions, which emphasized either ingroup cooperation (“cooperative”), intergroup competition (“competitive”), or individual performance (“no interaction”). The ideology-bias relation did not differ between conditions.

Finally, we conducted an internal meta-analysis to assess the overall effect size of ideological differences in intergroup bias across our four studies. This analysis revealed an average effect size across our studies of ($d = .12$, $SD = 0.03$, $z = 4.63$, 95% HPD CrI [0.055, 0.191], $BF_{10} = 18.70$). This effect was relatively small in size, but it was highly statistically significant according to conventional NHST criteria (stats), and our Bayes factor indicated strong evidence for the existence of ideological differences in bias.

Using these combined data, we also conducted more nuanced analyses that revealed several additional insights into the nature of the relation between ideology and intergroup bias. First, we found that this relation was primarily driven by ingroup favoritism, rather than outgroup derogation: On the whole, conservatives were more positive towards their ingroups than were liberals (stats)--for example, expressing greater identification and more positive evaluations. However, conservatives and liberals did not differ in their evaluations of their outgroups (stats; Fig 3). We also found that the strength of the relation between ideology and bias depended on how political ideology was measured (interaction stats): Our issue-based measure of political ideology was a stronger predictor of bias (stats) than was self-reported liberalism-conservatism (stats), although both measures exhibited positive relations with bias.

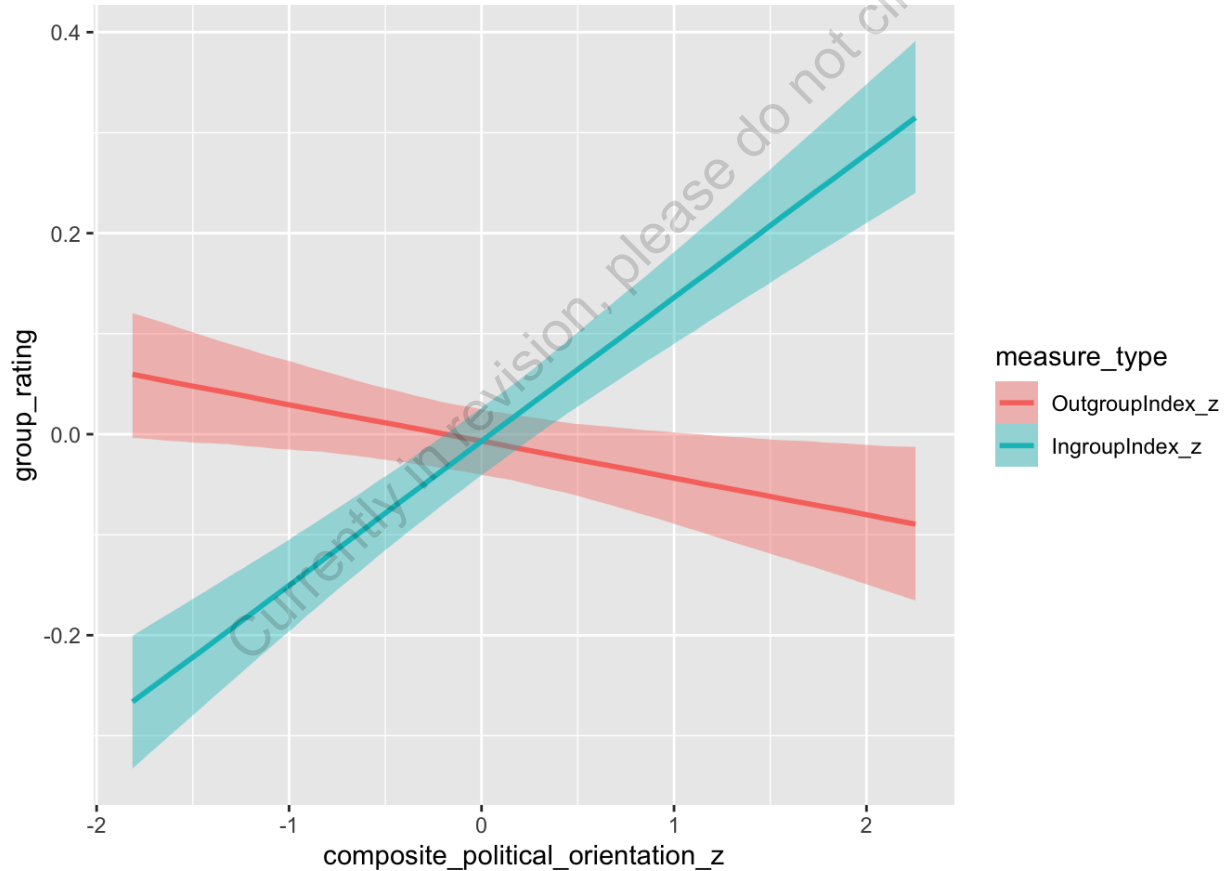


Figure 4. The relation between ideology and attitudes/behavior towards both ingroups and outgroups.

Finally, we also examined a related question that has also been the subject of considerable debate in the literature: whether ideological extremity—that is, the strength of an individual’s ideological identification—was related to intergroup bias. However, we found no evidence of a relation (NHST stats), and Bayesian analyses suggested that the evidence favored the null hypothesis of no relation (stats). As such, our results decisively demonstrated that political ideology—but not ideological extremity—is associated with intergroup bias.

Discussion

The results of these four studies advance the longstanding debate regarding whether liberals and conservatives differ in their propensity for intergroup bias. Our results decisively demonstrated that conservatives do in fact exhibit a stronger preference for their own versus other social groups. Although the magnitude of these ideological differences was relatively small, the effects were statistically significant and consistent across a variety of contexts, including different types of social groups (e.g., random versus volitional) and both cooperative and competitive intergroup contexts. Taken together, these findings lend strong support for the asymmetrical intergroup bias hypothesis, demonstrating that conservatives are more inclined to favor their own social groups over others across a wide range of domains.

Our results also provided more nuanced insights into the nature of this relation. In particular, our studies indicated that conservatives’ intergroup bias primarily manifests as increased warmth and identification with their own groups rather than outright hostility or negative stereotyping of outgroups. This pattern suggests that conservative bias is less about disliking others and more about elevating those with whom they share group membership. However, the practical implications of this bias remain important, as it still resulted in resource allocation and decision-making that systematically disadvantaged outgroups.

Importantly, however, our results also suggested that ideological differences in intergroup bias may be overridden when group membership is based on shared moral and political values: In Study 2, when participants were led to believe that their minimal groups were assigned based on similar moral and political attitudes, ideological differences in bias effectively disappeared. This result highlights the context-dependent nature of intergroup bias and suggests that shared values can override the tendency for conservatives to favor their own groups. This attenuation of bias in politically and morally aligned groups may also help to explain why, despite evidence of conservative bias in many domains, research on political polarization often finds symmetrical levels of mutual dislike between liberals and

conservatives. Our findings suggest that the political nature of these groups may neutralize the typical ideological differences in intergroup bias.

Importantly, however, our research leaves several questions unanswered. For instance, what drives the reduction of ideological differences in bias when groups are defined by shared values? While our data suggest that factors such as competition or volitional group choice do not account for this effect, other explanations remain to be explored. Future research could investigate whether factors like shared goals or perceived threats to group identity might play a role in diminishing ideological differences in intergroup bias.

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This finding is in keeping with past work showing that ideological differences in other traits and motivations can be overruled under particular circumstances, such as in political contexts (e.g., Stern & Ruisch, 2023; Ruisch et al., PLOS), as well as more broadly with the idea that shifting goal hierarchies can lead to contextual differences in when chronic motivations do and do not manifest in real-world behavior.

Methods

Ethics Information

All pilot studies conformed to ethical standards for research involving human participants and were approved by the Cornell University IRB (#1711007631). Our proposed studies were also approved by the Cornell University IRB (#1711007631). Participants will be recruited online, will provide informed consent, and will be compensated at a rate of \$0.20 per expected minute of participation.

Pilot Data

To provide an initial test of ideological (a)symmetries in minimal intergroup bias, we conducted a series of three pilot studies (total $N = 1,001$). In these studies, we examined intergroup bias under the most minimal of conditions, basing group assignment on either random (Pilots 1 and 2, random-number generator) or arbitrary (Pilot 3, overestimator/underestimator paradigm) characteristics, keeping the studies extremely short (2-3 minutes), and including no tasks other than group assignment, a measure of ingroup identification, and an allocation task. Our three studies were nearly identical save for the method used to assigned participants to their groups (random vs. arbitrary). Analyses revealed that neither the pattern of results nor effect sizes differed between these three studies (all $ps > .07$), and so in order to increase statistical power^{76,77} and provide a more accurate estimate of effect sizes⁷⁸, we collapsed across these studies and simply present the combined results. We provide a brief overview of the method and results below. (See Supplementary Information: Pilot Data Extended Methods and Results and Supplementary Figures 1-3 for more information.) All data, materials, syntax, and preregistration documentation are available at

https://osf.io/jxsgw/?view_only=1b1994774989419eac3be214fce974f7.

In these studies, participants were first assigned to their minimal groups. Afterwards, they completed an allocation task created for the purposes of this study, in which they distributed \$.50 between one randomly selected member of their ingroup and one randomly selected member of the outgroup. Following the allocation task, participants indicated their level of identification with their ingroup using a single-item measure of group identification: “I identify with [group name]”, measured on a 1 (*strongly disagree*) to 7 (*strongly agree*) scale^{63,79}. They next reported their age and gender and then indicated their political orientation on a 1 (*extremely liberal*) to 9 (*extremely conservative*) scale.

Providing tentative support for the asymmetrical intergroup bias hypothesis, more conservative individuals expressed significantly greater identification with their minimal ingroups ($\beta = .09$, $t(950) = 2.63$, $p = .009$, 95% CI[.02,.15]). A $BF_{10} = 2.14$, suggests that the alternative hypothesis that conservatism and ingroup identification are related is twice as likely than the null hypothesis that they are not. However, there was no significant relation between ideology and ingroup allocations ($\beta = .03$, $t(949) = 0.79$, $p = .43$, 95% CI[-.04,.09]), $BF_{10} = 0.10$.

In exploratory analyses, we also simultaneously assessed the predictive power of both ideology (liberal versus conservative) and ideological *extremity* (*degree of* liberalism-conservatism, operationalized as the distance from the “moderate” midpoint of the ideology scale⁸⁰). We found that ideology predicted greater ingroup identification ($\beta = .12$, $t(749) = 3.30$, $p = .001$), but we found no evidence for an association between ideological extremity and identification ($\beta = .08$, $t(749) = 1.73$, $p = .08$, 95% CI[-.01,.18]). Extremity was also not significantly related to allocations ($\beta = .04$, $t(749) = 1.12$, $p = .25$, 95% CI[-.04,.15]).

The results of these pilot studies therefore presented something of a mixed picture. On one hand, conservatism was significantly associated with greater ingroup identification. However, these ideological differences did not extend to our allocation measure. Further, the relation between ideology and ingroup identification, although statistically significant, was nonetheless quite small. (Importantly, the effect size of the relation between ideology and identification did not reach the $r = .1$ threshold that we define as our smallest effect size of interest in our proposed studies.) This may indicate that true ideological differences in ingroup favoritism are indeed quite small, or it may reflect other features of our study, such as (1) the particularly minimal nature of our research design, (2) the liberal skew of our participant sample, and/or (3) our use of a single-item measure of left-right ideology, rather than more comprehensive and multifaceted measures (e.g., issue-based scales and/or separate assessments of social and economic liberalism-conservatism). In our proposed research, we recruit a more representative participant sample, include a more comprehensive measure of political ideology, and use

a range of different paradigms in order to better understand the true nature and size of this effect, as well as whether it may be amplified or attenuated under different conditions.

Further, many other questions remain, such as whether and to what degree these ideological differences might manifest on other measures of intergroup bias—including more established measures of resource allocation (versus the unvalidated measure used in our pilot studies). Given that ingroup identification has been shown to be a powerful predictor of many forms of minimal ingroup favoritism^{61–63,71}, including intergroup bias on evaluative measures (e.g., warmth/liking and trait ratings), as well as established allocation measures (e.g., the Tajfel matrices), conservatives' greater identification with their ingroup suggests that they may exhibit other forms of ingroup favoritism as well. In the proposed research, below, we provide a more systematic examination of this question using a range of validated outcome measures.

Design

Studies 1A and 1B

In Studies 1A and 1B, we will build on our pilot studies in four central ways. First, we will conceptually replicate and extend the pilot results using more diverse samples (a sample that is distributed similarly to the adult US population on major demographic characteristics (age, gender and race/ethnicity), as well as a Mechanical Turk sample with an oversample of conservatives) in order to examine the generalizability of these effects. Second, we will use a more engaging and better-validated minimal groups paradigm (a longer form of the overestimator/underestimator task from Pilot 3). Because research and theory suggest that ideological differences in ingroup favoritism may emerge more strongly for social groups based on some kind of (ambiguously) meaningful shared trait²⁷, we anticipate that this paradigm will amplify ideological differences in minimal group favoritism. Third, we will include three distinct outcome measures: ingroup identification, trait evaluations, and an empirically validated allocation task. Fourth, we will use a more comprehensive measure of political ideology, which includes both an issue-based scale as well as separate assessments of social/cultural and economic liberalism-conservatism. This will give us a more complete understanding of how ideological differences in ingroup favoritism may manifest.

Our allocation measure in these studies will be the Tajfel matrices²¹, a measure that has been widely used and well-validated in past research^{18,20,53–55,81–84}. This measure asks participants to choose

between payoff matrices that award differing amounts of points (which correspond to different sums of money) to members of their ingroup and the outgroup. These matrices capture both individual differences in general ingroup favoritism in allocation, as well as more fine-grained distinctions among four types of intergroup reward strategies: Maximum Ingroup Profit (MIP; giving the maximum amount to the ingroup, regardless of the amount awarded to the outgroup); Parity (P; distributing equal amounts to both groups); Maximum Differentiation (MD; maximizing the positive difference in points between ingroup and outgroup members); and Maximum Joint Profit (MJP; maximizing the total amount distributed, independent of the target group^{21,82}).

The design of Studies 1A and 1B will be identical save for the specific participant sample used. This will allow us to more conclusively determine whether these two samples meaningfully differ, as well as to analyze the combined dataset to gain additional statistical power to detect small effects.

Procedure. After providing informed consent, participants will be assigned to their minimal groups based on the dot estimation paradigm used in the original minimal groups studies^{19,21} and much subsequent research^{18,55}. Participants will first complete a task in which they will view images consisting of random constellations of small black dots and will be asked to estimate the number of dots in each image. Each image will be presented for 300 milliseconds, after which participants will make their estimate. There will be 10 images in total. After the estimation task, participants will be informed about the (ostensible) nature of the groups. They will be told that past research has found that when estimating quantities, some people tend to consistently underestimate, while others tend to consistently overestimate. They will be informed that although these two groups do not differ in their degree of accuracy in these tasks, these individual differences appear to be relatively stable, and researchers suspect that they may correlate with other aspects of personality as well.

After receiving this information, participants will be informed of the group to which they belong, either Underestimators or Overestimators (group assignment will in fact be random). Participants will then complete our three dependent measures, which will be presented in random order:

Evaluative Trait Ratings: Participants will be asked to provide their trait ratings of Underestimators and Overestimators, ostensibly for the purpose of assessing lay perceptions of how the two groups might differ.⁸⁵ To this end, participants will be asked to rate both groups on a number of traits. There will be 6 traits in total, three negative (lazy, selfish, incompetent) and three positive (able, intelligent, kind). Participants will be asked to rate how they think that both Underestimators and Overestimators would score on each trait, using 7-point semantic differential scales.

Ingroup Identification: Participants will report their identification with both the ingroup and the outgroup by endorsing agreement with two statements of identification, one for each group⁷⁹.

Tajfel Matrices Allocation Task: Participants will complete a task in which they will (ostensibly) assign points to other randomly selected participants from the study, one ingroup member and one outgroup member. The other individuals will be identified only by an anonymous identification number. Each point in the task will be said to correspond to \$.10, to be distributed at the end of the study. Participants will be presented with the six Tajfel matrices that have been most commonly used in past research, consisting of different relative ingroup/outgroup allocations⁸². For each matrix, participants will be asked to choose the distribution of points that they prefer (the first pair of matrices compares Ingroup Favoritism (FAV = Maximum Ingroup Profit + Maximum Differentiation) versus Parity (P; equal amounts to ingroup and outgroup); the second pair compares FAV versus Maximum Joint Profit (MJP), and the third pair compares Maximum Differentiation (MD) against Maximum Ingroup Profit (MIP) + Maximum Joint Profit (MJP). (See Supplementary Information: Measures and Materials for Proposed Studies for more detail.)

Participants will then indicate their political orientation using (1) the same item from our pilot studies assessing global liberal-conservative self-identification, (2) a single-item self-identification measure of social and cultural liberalism/conservatism (“In terms of social and cultural issues, how liberal or conservative are you?”), (3) a single-item self-identification measure of economic liberalism/conservatism (“In terms of economic issues, how liberal or conservative are you?”), and (4) an issue-based measure of political ideology (the 12-Item Social and Economic Conservatism Scale; SECS⁸⁶), which assesses participants’ degree of liberalism/conservatism based on their ratings of 12 political issues/principles (e.g., welfare benefits, gun ownership, abortion rights). Participants will then provide demographic information. (See Supplementary Information: Measures and Materials for Proposed Studies for more detail.)

Based on recent best practice recommendations for conducting studies with online samples⁸⁷, in these and all subsequent studies we will take steps to ensure good data quality and avoid potential “bots” (i.e., automated computer programs that mimic human behavior). Specifically, we will prevent participants from completing more than one study, and we will include a “CAPTCHA” verification question that will require participants to click a box marked “I am not a robot” to continue with the study, a common method of preventing bot traffic. For this and all remaining studies, we will also include the following instructional attention/comprehension check⁸⁸: “What color is the sky? For this question, please select purple to confirm you are reading the questions carefully.”, with the response

options “blue”, “red”, “green”, “purple”, “white”, “yellow”, “gold”, “orange”, and “black”. We will exclude participants who fail this check (i.e., those who do not select “purple”) and replace them in our samples.

Study 2

In Study 2, we will build on the previous results by examining potential moderators and mediators of the relation between intergroup bias and ideology to make two central contributions. First, we will manipulate the nature of the minimal groups—whether they are formed randomly, by volition, or based on (ostensible) attitudinal similarity—in order to determine whether group type moderates the size of any observed association between ideology and ingroup favoritism.

Based on past research, which tends to suggest that bias and motivated cognition often flourish most under conditions of ambiguity^{89,90}, we anticipate that groups based on clearly and explicitly random criteria would tend to elicit little intergroup bias among liberals and conservatives alike. That is, when group assignment is explicitly and unambiguously *not* meaningful, there should be minimal “psychological leeway” for either ideological group to engage in intergroup bias. However, for groups based on relatively more (yet still somewhat ambiguously) meaningful groups (e.g., people that chose to belong to “the Green Team” or scored similarly on short personality/belief measures), it should be that case that bias—and ideological differences in bias—should emerge more strongly.

Second, we will examine factors that may statistically account for any observed associations between ideology and intergroup bias, drawing from several influential theories of conservatism. We will focus on three possibilities:

(1) Ingroup-protective moral values. Work on moral foundations theory has argued that conservatives more strongly endorse ingroup-protective moral values, such as loyalty towards one’s own social groups^{8,91}. To the degree that this is true^{38,40}, these moral values may lead conservatives to show greater favoritism to their minimal ingroups as well.

(2) Intolerance of uncertainty. One leading theory regarding the nature of the left-right divide is that political conservatism is related to fundamental psychological needs, such as epistemic needs to avoid uncertainty^{33,92}. Given that uncertainty avoidance has been shown to lead individuals to more strongly identify with their social groups—including minimal groups^{93,94}—ideological differences in motivations to resolve or avoid uncertainty could also help explain conservatives’ greater identification with their minimal groups.

(3) Social norms. Perceived social norms regarding the acceptability of ingroup favoritism play a strong role in shaping discrimination, both in minimal groups⁹⁵ and in the real world⁵². Past research has

shown that many conservatives tend to view modern society as a competitive, “dog-eat-dog” world, expecting others to be more aggressive and to primarily look out for their own interests⁹⁶. These findings suggest that conservatives may see outgroup derogation and ingroup favoritism as more normatively common and/or acceptable intergroup interaction strategies, which may shape their own responses in these tasks⁴³.

Procedure. After providing informed consent, participants will complete the three measures assessing the three potentially related psychological processes discussed above: (1) the three “binding foundation” subscales (loyalty, respect for authority, and purity) of the 30-item moral foundations questionnaire⁹⁷; (2) the 15-item need for closure scale (a measure of intolerance of uncertainty⁹⁸), and four questions assessing perceptions of the descriptive and prescriptive norms surrounding ingroup favoritism and intergroup bias (adapted from Paluck & Shepherd⁹⁹). They will also complete the measures of political ideology from the previous studies and provide demographic information.

Participants will then be randomly assigned to one of three conditions, which will determine the method by which their minimal groups will be assigned. Participants in the Attitude-Similarity Group Condition will be assigned to either “The Blue Group” or “The Green Group,” ostensibly based on their responses to the preceding measures (group assignment will in fact be random; adapted from ⁶³). Participants in the Volitional Group Condition will be asked to choose the group to which they wish to belong, either Blue or Green (adapted from ⁷²). Participants in the Random Group Condition will be asked to click a button on their computer screen in order to generate a random number. Based on the number they receive, participants will be assigned to either The Blue Group or The Green Group. Participants will then complete the three dependent measures from Studies 1A and 1B: ingroup identification, intergroup evaluations, and intergroup allocations, presented in random order. We will include the same quality check question and exclusion criteria as in Study 1.

Study 3

In our third study, we will build on our previous findings by examining an additional potential moderator: whether ideological differences in intergroup bias differ as a function of whether the intergroup interaction is cooperative versus competitive. That is, theories of conservatism and related belief systems have argued that ideological beliefs are largely situationally dependent, becoming more salient or “activated” under conditions of intergroup competition or threat, and that they exert a stronger influence on cognition and behavior under these circumstances^{72,100}. We therefore predict that ideological differences in intergroup bias may emerge more strongly under competitive (versus cooperative) intergroup conditions.

Procedure. Upon beginning the survey, participants will be informed that there will be a short wait while other individuals log in. They will be placed in a “virtual waiting room,” which will graphically depict the virtual entry of other participants into the study, by slowly adding silhouettes depicting other “players” to the screen (intended to bolster the cover story that participants are playing in real time with other individuals). After approximately 30 seconds, the study will begin.

After providing informed consent, participants will be assigned to their minimal groups. They will click a button on the screen in order to generate a random number. Based on this number, they will then be assigned to either the “The Red Group” or “The Green Group.” All participants will then be informed that they will be completing a problem-solving task, and they will be shown three sample problems (mental rotation problems^{101,102}) to familiarize them with the task. Participants will then receive further information about the structure of the problem-solving task, which will differ between participants by random assignment. Participants in the *No Interaction Condition* will be told that each person in the study will complete the problem-solving task independently and will be judged solely on the basis of their own work. They will be informed that any participant that meets the threshold will receive a \$0.50 bonus. Participants in the *Cooperative Interaction Condition* will be informed that they will work together with the other members of their team in the problem-solving task. They will be informed that the scores of each member of their group will be added together, and that all group members will receive a \$0.50 bonus if their group’s combined score meets the threshold. Participants in the *Competitive Interaction Condition* will be told that their group will compete against the opposing group in the problem-solving task. They will be informed that the scores of each group member will be added together in order to determine which group received the higher overall score, and that each member of the winning group will receive a \$0.50 bonus.

Participants will then complete the mental rotation task. Following this, they will complete the three dependent measures from our previous studies, measuring ingroup identification, intergroup evaluations, and intergroup allocations. They will then indicate their political orientation and provide demographic information. We will include the same quality check question and exclusion criteria as in Study 1. (See Supplementary Information: Measures and Materials for Proposed Studies for more detail.)

Study 4

We will conduct a final study to replicate and extend the previous studies by examining whether ideological differences in intergroup bias—under whatever conditions, if any, they may exist—extend to

an *implicit* measure of minimal group favoritism. (In other words, the purpose of this study is not to determine the strength of the relation between ideology and intergroup bias, but rather to determine whether such associations, if and when they exist, may extend to more implicit processes.) Finding ideological differences on an implicit measure would lend support to the perspective that ideological differences in intergroup favoritism may unfold relatively automatically and uncontrollably, perhaps stemming from low-level cognitive mechanisms that shape the importance and/or salience of the psychological distinction between “us” and “them”^{27–31}. Conversely, finding little or no ideological differences on implicit measures of ingroup identification would lend support to the argument that liberals and conservatives do not differ in their degree of implicit positivity/negativity towards social ingroups and outgroups, and that ideological differences in prejudice and discrimination—to the degree that they exist—likely arise from more deliberative processes relating to attitudes towards engaging in explicitly discriminatory behavior⁴³. From a practical perspective, understanding the degree to which ideological differences in intergroup bias are relatively automatic versus deliberative in nature may provide insight for the development of new methods of reducing intergroup bias, given their differing effects on intergroup behavior¹⁰³, dissociable mechanisms of updating^{104,105}, and vulnerability to self-presentation motives¹⁰⁶.

Procedure. For this study, we will use the previous minimal-group assignment paradigm (i.e., overestimator/underestimator, random groups, or attitude similarity) that shows the strongest relation between political ideology and intergroup bias. Following the group assignment manipulation, participants will complete a task in which they will be asked to familiarize themselves with the names of the individuals from both groups. They will be shown two lists of five names, ostensibly the names of other individuals who belong to their ingroup and outgroup, randomly selected from the following list: Andrew, Lindsay, Anthony, Susan, William, Amy, Steven, Tara, Adam, and Courtney. To best reflect the likely demographics of our sample of typical Americans, this list was drawn randomly from the top 50 most common male and female names for people born in the 1970s, 1980s, and 1990s (US Social Security Administration: <https://www.ssa.gov/oact/babynames/decades/index.html>). Participants will be given a few moments to look over the list, and then will complete a computer-based task in which they will memorize the names of the members of each group, following established procedure¹⁰⁷. The task will consist of two blocks of 30 trials in which a single name will appear on the screen, and participants will be asked to assign it to the appropriate group.

Following this task, participants will then complete an Implicit Association Test (IAT¹⁰⁸), designed to measure their implicit identification with their minimal group¹⁰⁷. The IAT categories will be represented

by the group names (e.g., “Red” and “Green”; “Overestimator” and “Underestimator”) and the stimuli for the trials will be the team members’ names. The attribute categories will be “self” (I, me, mine, my, self) and “other” (other, their, theirs, them, they). Response latencies will be calculated following established recommendations¹⁰⁸, such that higher values indicate greater implicit identification with the ingroup (versus outgroup). Also following these recommendations, we will omit individual trials with latencies greater than 10,000ms, and we will exclude participants who have response latencies of less than 300ms on more than 10% of the IAT trials. Finally, we will also exclude participants who press the same keyboard key throughout the entire IAT. Following the IAT, participants will complete the (explicit) ingroup identification, intergroup evaluation, and intergroup allocation measures from Studies 1-3. Finally, they will indicate their political orientation and provide demographic information. We will include the same quality check question and exclusion criteria as in Study 1.

Sampling Plan

In all studies, we define our “smallest effect size of interest”¹⁰⁹ as a correlation of $r = .10$, following past research that has argued that smaller effects are unlikely to have meaningful real-world impact¹¹⁰. In line with these arguments, we anticipate that this is the smallest effect that could realistically be expected to lead to practically significant ideological differences in intergroup attitudes or behavior in the real world. We derived our minimum target sample sizes for all studies based on power analyses for 95% power to detect an effect of $r = .10$ (one-tailed test).

Studies 1A and 1B

Based on the power analysis described above (95% power for an effect of $r = .10$), for Studies 1A and 1B we set a target sample size (after exclusions) of 1,073 participants for each study. We will exclude participants who fail our included attention/comprehension check (i.e., those who do not select “purple” as instructed) and then replace these participants in our samples to ensure that we reach our target sample size and statistical power. For Study 1A, we will recruit a sample of participants that is distributed similarly to the US adult population on the basis of age, gender and race/ethnicity using Prolific Academic’s panel service. This is an opt-in sample that is matched to census data distributions of the U.S. adult population on demographic factors (age, gender, race/ethnicity).

For Study 1B, we will recruit a Mechanical Turk sample with a 15% oversample of conservatives. Based on the distribution of ideology in our pilot studies and other recent MTurk studies, this should provide us with an approximately equal number of liberals and conservatives and allow us to

counterbalance the liberal skew of typical MTurk populations to obtain a more ideologically diverse sample. To recruit this conservative oversample, we will use CloudResearch's (cloudresearch.com) MTurk toolkit. The MTurk toolkit allows researchers to specifically target MTurk workers who have particular characteristics—in this case, those who identify as politically conservative—based on their answers to CloudResearch's screening surveys. (Importantly, qualifying participants are unaware that they are eligible for the study based on possessing a specific characteristic. Rather, the program operates by making the study unavailable to individuals who do not possess this characteristic, ensuring that participants' responses are not biased by preexisting knowledge of the traits that are of interest for a particular study.) Our conservative oversample will consist of individuals who identified as either "conservative" or "very conservative" in a previous screening survey.

Studies 2 and 3

We will recruit a sample of 4,854 participants after exclusions (1,618 per condition, based on an anticipated between-condition *SD* ratio of 1:1.1) per study. This will allow us to detect condition differences in linear regression slopes of .1 at 95% power and provide us with 99.2% power to detect our smallest effect of interest ($r = .10$) *within* each experimental condition (GPower 3.1¹¹¹).

If sample type moderates the relation between ideology and intergroup bias in Studies 1A and 1B (see analysis plan below), we will recruit for Studies 2 and 3 whichever participant sample showed the strongest association between ideology and intergroup favoritism (to enhance our ability to examine the processes(s) related to this effect). If no sample differences are observed, we will recruit a Mechanical Turk sample with an oversample of conservatives for Studies 2 and 3, as in Study 1B.

Study 4

As noted above, a power analysis for 95% power to detect an effect of $r = .10$ recommends a sample size of 1,073, after exclusions. The sample will be recruited from the same source (either Prolific Academic's panel service or Mechanical Turk as described above) as the study that we are replicating and extending.

Analysis Plan

All Studies

Data collection will be automated by Prolific and Mechanical Turk with participants randomly assigned to each condition and experimenters will thus be blind to this assignment. All analysis scripts will be prepared using sample data in advance of data collection so that analysis can proceed with

experimenters blind to the individual data of each participant (e.g., condition, assignment, political ideology, etc.).

Calculation of Independent and Dependent Variables

Calculation of political ideology: To assess political ideology, we will first recode the reverse-scored items from the issue-based SECS so that higher scores indicate greater conservatism. We will then calculate mean scores on this measure. We will then create a composite index of political liberalism/conservatism by z-scoring and calculating the mean of the SECS, the global political orientation item, the social/cultural conservatism item, and the economic conservatism item. This composite score will serve as our measure of ideology in all analyses.

Calculation of intergroup bias: Ingroup identification will be calculated by subtracting the difference between participants' rated identification with their outgroup from their rated identification with their ingroup. Evaluative intergroup bias will be defined as the difference between mean positivity towards the ingroup and mean positivity towards the outgroup on the trait ratings (with negative trait ratings reversed scored). Finally, a simple difference score between the number of points allocated to the ingroup minus the number allocated to the outgroup¹⁸ will serve as our primary measure of intergroup bias. Additionally, in exploratory analyses we will calculate "pull scores" to quantify the relative preferences for each of the intergroup strategies described above (MIP, P, MJD, and MD), following the procedure outlined by Brown and Bourhis^{82,112,113}.

Finally, we will compute a composite intergroup bias index for use in our primary tests of our hypotheses. To do so, we will standardize the identification, evaluation, and allocation measures described above. The mean of these three measures will serve as our primary dependent measure for these analyses.

Reliability: For all scale measures, we specify the minimum necessary reliability score (alpha) as .70. For any scales for which this alpha is not achieved, we will conduct a factor analysis to identify subscales that exhibit sufficient reliability, following the same .70 cutoff. We will then separately report the relation between these subscales and our variables of interest in the Supplemental Information.

Main Hypothesis Testing (see Design Table)

The existing literature (and our pilot data) does not provide clear reasons to predict that liberals may generally exhibit greater intergroup bias—rather, the competing predictions in the literature, between which we aim to adjudicate, are (1) that conservatives exhibit greater intergroup bias, or (2) there are no ideological differences in intergroup bias. Given this, we will employ one-tailed statistical tests for examining ideological differences in intergroup bias. Nonetheless, we include a plan for how we

will interpret the full set of possible outcomes of our statistical tests. Below, we describe the specific results that would constitute support for each hypothesis. Results that are not statistically significant or which are significant but with effect sizes below our smallest size of interest will undergo inferiority testing, as described below.

Hypothesis: Political conservatism will be associated with intergroup bias. We will use linear regression to test this hypothesis. Political ideology (the continuous composite measure) will be entered as the independent variable, and our composite intergroup bias measure will be entered as the dependent variable. We will consider this hypothesis supported if:

- (a) this relationship is positive (such that greater conservatism is associated with greater intergroup bias) and statistically significant at $p = .05$ or below, and
- (b) the size of this association is $r = .10$ or above, our smallest effect size of interest.

Inferiority testing: To interpret potential null results, we will conduct an inferiority test if our measure of intergroup bias either shows (1) a non-significant association with ideology and/or (2) an effect size of less than $r = .10$. The inferiority test operates as an equivalence test with the lower bound set to infinity. It therefore allows us to determine whether we can reject the null hypothesis that the effect is at least as large as $r = .10$ and examine the probability that—given an observed effect size—there exists a true effect in the population that is at least as large as the *smallest effect size of interest (SESOI)*^{109, 75}. To conduct these analyses, we will calculate the 90% confidence interval around the observed correlation between ideology and intergroup bias. If this interval does not include (or exceed) .1, we will conclude that the effect of ideology on intergroup bias either does not exist or is too small to be of any practical importance¹⁰⁹. For example, should we find a positive but small relationship between ideology and intergroup bias of $r = .05$ and the 90% CI of this effect does not include (or exceed) .1, we will interpret this as evidence against the intergroup asymmetry hypothesis.

Bayes Factors: In addition to the NHST-based statistical approach outlined above, we will also compute, report, and interpret Bayes factors for each of our primary analyses of interest in line with the recommendations of Lee and Wagenmakers¹¹⁴. This will allow us to assess the strength of the evidence in supporting either the asymmetrical intergroup bias hypothesis or the null hypothesis (i.e., the symmetrical intergroup bias hypothesis).

Exploratory Analyses: Ideological Extremity

In all studies, we will also conduct exploratory analyses to determine whether ideological extremity (that is, a person's degree of liberalism/conservatism, from moderate to extreme) predicts intergroup bias. To assess ideological extremity, we will calculate the absolute distance between the midpoint of the scale and participants' responses for each of the four measures of political ideology listed above.⁸⁰ We will then z-score these items and compute the mean to create a single index of ideological extremity. This will serve as our measure of extremity in all analyses.

Studies 1A and 1B

Moderation Analyses

We will examine whether the size or nature of the association between ideology and intergroup bias differs meaningfully as a function of the specific sample—i.e., the national sample versus the MTurk sample with conservative oversample. We will combine the two datasets, adding sample type as a categorical moderator variable. We will then examine the interaction between sample type and ideology on our composite measure of intergroup bias. If the interaction is significant at $p < .05$ we will conduct simple slopes analyses to examine to what degree the size and/or direction of the effect differs between samples.

We will also conduct a series of exploratory moderation analyses (see Supplementary Information: Exploratory Analyses). In one model, we will separately examine the components of the intergroup bias difference scores (e.g., points assigned to the ingroup and points assigned to the outgroup), to examine whether ideology is more strongly associated with ingroup favoritism or outgroup derogation. In separate exploratory analysis, we will also examine the interaction between ideology in predicting minimal intergroup bias using the composite intergroup bias measure.

Study 2

Moderation Analyses

Ideology X group type interaction: We will examine whether minimal group type (random versus volitional versus attitude similarity) moderates the size of the relationship between ideology and ingroup favoritism using a linear regression model with dummy-coded variables for condition (for volitional group and attitude similarity group), ideology, and their interactions as predictors of intergroup bias. If the interactions are non-significant, and/or effect sizes are smaller than $r = .10$, we will conduct inferiority tests following the procedures above.

Mediation Analysis

To test for mediation, we will use the PROCESS macro for SPSS¹¹⁵, a statistical package for computing mediation and moderation models that uses a path analysis framework^{116,117}. For continuous outcomes (such as our measures of intergroup bias) in a mediation model, the PROCESS Macro estimates the coefficients of the model using OLS regression and generates both direct and indirect effects. The PROCESS Macro also provides bias corrected and percentile based bootstrap confidence intervals for conditional and unconditional indirect effects.

We will conduct a mediation analysis in which our composite measure of political orientation will be entered as the predictor (X) variable, our composite measure of intergroup bias will be entered as the dependent (Y) variable, and our three proposed mediator variables (ingroup-protective morality, intolerance of uncertainty, and perceived norms) will be entered as the mediator (M) variables. This will give us 95% power to detect mediation assuming correlations of .1 or greater between the predictor/outcome/mediators within each condition with $ns = 1618$ and $> 99\%$ power across conditions with our total $N = 4,854$ ¹¹⁸.

Study 3

Moderation Analyses

The ideology X group type interaction will be examined following the procedure outlined above in Study 2.

Study 4

Implicit Ingroup Favoritism

We will test whether ideology is associated with greater implicit ingroup identification. This analysis will be parallel to that outlined in “Main Hypothesis Testing” but IAT D-score will be the dependent variable.

Summary Analyses

Following recent methodological recommendations⁷⁷, we will also conduct an internal “mini-meta-analysis” of these five studies to provide additional statistical power, estimate an average effect size (across different participant samples and conditions)^{76,78}, and examine potential moderators. Because we will have a nested structure, with individual studies having multiple distinct groups of participants (e.g., those assigned to different conditions in Study 2), and all studies collecting multiple observations (i.e., distinct measures of intergroup bias and political ideology) from a single group of participants, we will fit a multi-level meta-analysis model, specifying nested random effects for study, group, and observation¹¹⁹. We will compute the average effect size of the relations between ideology and intergroup bias. We will conduct Cochran’s Q-tests to determine if there is significant heterogeneity in

our observed effect sizes. If so, we will examine potential moderators of this effect (e.g., sample type: MTurk versus Prolific; Different group conditions as in Studies 2 and 4).

In exploratory analyses, we will also compute average effect sizes for each of the individual dimensions of our independent (e.g., self-reported ideology, issue-based ideology) and dependent measures (e.g., intergroup evaluations, intergroup allocations) to determine whether the ideology-bias relation differs as a function of the specific measures that are used.

As above, for all analyses in which statistically significant results are not obtained and/or average effect sizes are smaller than $r = .10$, we will conduct inferiority tests by computing 90% confidence intervals. If the resulting confidence intervals do not include (or exceed) .1, we will conclude that the effect (e.g., of ideology on that specific measure of ingroup favoritism) either does not exist or is too small to be of any practical importance¹⁰⁹.

Data Availability

We commit to sharing all of our raw data and materials through our OSF site for the project, [.](#)

Code Availability

We commit to sharing all of our code through our OSF site for the project,

https://osf.io/jxsgw/?view_only=1b1994774989419eac3be214fce974f7.

References

1. McCoy, J., Rahman, T. & Somer, M. Polarization and the Global Crisis of Democracy: Common Patterns, Dynamics, and Pernicious Consequences for Democratic Polities. *Am. Behav. Sci.* **62**, 16–42 (2018).
2. Pew Research Center. Political Polarization in the American Public. (2014).
3. Pew Research Center. Partisanship and political animosity in 2016. (2016).
4. Pew Research Center. The partisan divide on political values grows even wider. (2017).
5. Reiljan, A. ‘Fear and loathing across party lines’ (also) in Europe: Affective polarisation in European party systems. *Eur. J. Polit. Res.* **59**, 376–396 (2020).
6. Iyengar, S. & Westwood, S. J. Fear and Loathing across Party Lines: New Evidence on Group Polarization. *Am. J. Polit. Sci.* **59**, 690–707 (2015).
7. Pew Research Center. Political Typology Reveals Deep Fissures on the Right and Left. (2017).
8. Graham, J., Haidt, J. & Nosek, B. A. Liberals and conservatives rely on different sets of moral foundations. *J. Pers. Soc. Psychol.* **96**, 1029–1046 (2009).
9. Terrizzi, J. A., Shook, N. J. & McDaniel, M. A. The behavioral immune system and social conservatism: a meta-analysis. *Evol. Hum. Behav.* **34**, 99–108 (2013).
10. Thornhill, R. & Fincher, C. What is the relevance of attachment and life history to political values? *Evol. Hum. Behav.* **28**, 215–222 (2007).
11. Altemeyer, B. *Right-wing authoritarianism*. (University of Manitoba press, 1981).
12. Altemeyer, B. *Enemies of freedom: Understanding right-wing authoritarianism*. (Jossey-Bass, 1988).

13. Duckitt, J. A dual-process cognitive-motivational theory of ideology and prejudice. in *Advances in Experimental Social Psychology* vol. 33 41–113 (Elsevier, 2001).
14. Hodson, G. & Busseri, M. A. Bright Minds and Dark Attitudes: Lower Cognitive Ability Predicts Greater Prejudice Through Right-Wing Ideology and Low Intergroup Contact. *Psychol. Sci.* **23**, 187–195 (2012).
15. Brandt, M. J., Reyna, C., Chambers, J. R., Crawford, J. T. & Wetherell, G. The Ideological-Conflict Hypothesis: Intolerance Among Both Liberals and Conservatives. *Curr. Dir. Psychol. Sci.* **23**, 27–34 (2014).
16. Chambers, J. R., Schlenker, B. R. & Collisson, B. Ideology and Prejudice: The Role of Value Conflicts. *Psychol. Sci.* **24**, 140–149 (2013).
17. Crawford, J. T., Brandt, M. J., Inbar, Y., Chambers, J. R. & Motyl, M. Social and economic ideologies differentially predict prejudice across the political spectrum, but social issues are most divisive. *J. Pers. Soc. Psychol.* **112**, 383–412 (2017).
18. Brewer, M. B. In-group bias in the minimal intergroup situation: A cognitive-motivational analysis. *Psychol. Bull.* **86**, 307–324 (1979).
19. Tajfel, H. Experiments in intergroup discrimination. *Sci. Am.* **223**, 96–103 (1970).
20. Spears, R. & Otten, S. Discrimination: revisiting Tajfel's minimal group studies (chapter 10). in *Social psychology: Revisiting the classic studies* (eds. Smith, J. R. & Haslam, S. A.) (SAGE, 2012).
21. Tajfel, H., Billig, M. G., Bundy, R. P. & Flament, C. Social categorization and intergroup behaviour. *Eur. J. Soc. Psychol.* **1**, 149–178 (1971).
22. Adorno, T. W., Frenkel-Brunswik, E., Levinson, D. & Sanford, N. *The authoritarian personality*. (Harper, 1950).

23. Pratto, F., Sidanius, J., Stallworth, L. M. & Malle, B. F. Social dominance orientation: A personality variable predicting social and political attitudes. *J. Pers. Soc. Psychol.* **67**, 741–763 (1994).
24. Sidanius, J. & Pratto, F. The inevitability of oppression and the dynamics of social dominance. in *Prejudice, politics, and the American dilemma*. 173–211 (Stanford University Press, 1993).
25. Wilson, G. D. A dynamic theory of conservatism. in *The psychology of conservatism*. xv, 277–xv, 277 (Academic Press, 1973).
26. Haidt, J. The new synthesis in moral psychology. *Science* **316**, 998–1002 (2007).
27. Haidt, J. *The righteous mind: why good people are divided by politics and religion*. (Pantheon Books, 2012).
28. Brown, G. D. A., Fincher, C. L. & Walasek, L. Personality, Parasites, Political Attitudes, and Cooperation: A Model of How Infection Prevalence Influences Openness and Social Group Formation. *Top. Cogn. Sci.* **8**, 98–117 (2016).
29. Kessler, T. & Cohrs, J. C. The evolution of authoritarian processes: Fostering cooperation in large-scale groups. *Group Dyn. Theory Res. Pract.* **12**, 73–84 (2008).
30. Sinn, J. S. & Hayes, M. W. Replacing the Moral Foundations: An Evolutionary-Coalitional Theory of Liberal-Conservative Differences: Replacing the Moral Foundations. *Polit. Psychol.* **38**, 1043–1064 (2017).
31. Tuschman, A. *Our political nature: The evolutionary origins of what divides us*. (Prometheus Books, 2013).
32. Duckitt, J. & Sibley, C. G. A Dual-Process Motivational Model of Ideology, Politics, and Prejudice. *Psychol. Inq.* **20**, 98–109 (2009).

33. Jost, J. T., Glaser, J., Kruglanski, A. W. & Sulloway, F. J. Political conservatism as motivated social cognition. *Psychol. Bull.* **129**, 339–375 (2003).
34. Wilson, M. S. & Sibley, C. G. Social Dominance Orientation and Right-Wing Authoritarianism: Additive and Interactive Effects on Political Conservatism. *Polit. Psychol.* **34**, 277–284 (2013).
35. Duckitt, J. & Sibley, C. G. Right wing authoritarianism, social dominance orientation and the dimensions of generalized prejudice. *Eur. J. Personal.* **21**, 113–130 (2007).
36. Duckitt, J. & Sibley, C. G. The Dual Process Motivational Model of Ideology and Prejudice. in *The Cambridge Handbook of the Psychology of Prejudice* (eds. Sibley, C. G. & Barlow, F. K.) 188–221 (Cambridge University Press, 2017).
doi:10.1017/9781316161579.009.
37. Schein, C., Goranson, A. & Gray, K. The uncensored truth about morality. *The Psychologist* **28**, 982–985 (2015).
38. Schein, C. & Gray, K. The Unifying Moral Dyad: Liberals and Conservatives Share the Same Harm-Based Moral Template. *Pers. Soc. Psychol. Bull.* **41**, 1147–1163 (2015).
39. Schein, C. & Gray, K. The Theory of Dyadic Morality: Reinventing Moral Judgment by Redefining Harm. *Personal. Soc. Psychol. Rev.* **22**, 32–70 (2018).
40. Voelkel, J. G. & Brandt, M. J. The Effect of Ideological Identification on the Endorsement of Moral Values Depends on the Target Group. *Pers. Soc. Psychol. Bull.* **45**, 851–863 (2019).
41. Brandt, M. J. Predicting Ideological Prejudice. *Psychol. Sci.* **28**, 713–722 (2017).
42. Crawford, J. T. & Brandt, M. J. Ideological (A)symmetries in prejudice and intergroup bias. *Curr. Opin. Behav. Sci.* **34**, 40–45 (2020).

43. Wetherell, G. A., Brandt, M. J. & Reyna, C. Discrimination Across the Ideological Divide: The Role of Value Violations and Abstract Values in Discrimination by Liberals and Conservatives. *Soc. Psychol. Personal. Sci.* **4**, 658–667 (2013).
44. Lambert, A. J. & Chasteen, A. L. Perceptions of Disadvantage Versus Conventionality: Political Values and Attitudes Toward the Elderly Versus Blacks. *Pers. Soc. Psychol. Bull.* **23**, 469–481 (1997).
45. McClosky, H. & Chong, D. Similarities and Differences Between Left-Wing and Right-Wing Radicals. *Br. J. Polit. Sci.* **15**, 329–363 (1985).
46. Morgan, G. S., Mullen, E. & Skitka, L. J. When Values and Attributions Collide: Liberals' and Conservatives' Values Motivate Attributions for Alleged Misdeeds. *Pers. Soc. Psychol. Bull.* **36**, 1241–1254 (2010).
47. DellaPosta, D., Shi, Y. & Macy, M. Why Do Liberals Drink Lattes? *Am. J. Sociol.* **120**, 1473–1511 (2015).
48. Pew Research Center. Wide gender gap, growing educational divide in voters' party identification. (2018).
49. Bailey, A. H., LaFrance, M. & Dovidio, J. F. Is Man the Measure of All Things? A Social Cognitive Account of Androcentrism. *Personal. Soc. Psychol. Rev.* **23**, 307–331 (2019).
50. Bonilla-Silva, E. The invisible weight of whiteness: the racial grammar of everyday life in contemporary America. *Ethn. Racial Stud.* 1–22 (2011)
doi:10.1080/01419870.2011.613997.
51. Crandall, C. S., Eshleman, A. & O'Brien, L. Social norms and the expression and suppression of prejudice: The struggle for internalization. *J. Pers. Soc. Psychol.* **82**, 359–378 (2002).

52. Crandall, C. S. & Eshleman, A. A justification-suppression model of the expression and experience of prejudice. *Psychol. Bull.* **129**, 414–446 (2003).
53. Bourhis, R., Gagnon, A. & Sachdev, I. Les matrices de Tajfel: Un guide méthodologique pour la recherche intergroupes. *Cah. Int. Psychol. Soc.* **34**, 11–28 (1997).
54. Diehl, M. The Minimal Group Paradigm: Theoretical Explanations and Empirical Findings. *Eur. Rev. Soc. Psychol.* **1**, 263–292 (1990).
55. Otten, S. The Minimal Group Paradigm and its maximal impact in research on social categorization. *Curr. Opin. Psychol.* **11**, 85–89 (2016).
56. Lemyre, L. & Smith, P. M. Intergroup discrimination and self-esteem in the minimal group paradigm. *J. Pers. Soc. Psychol.* **49**, 660–670 (1985).
57. Turner, J. C. Social comparison and social identity: Some prospects for intergroup behaviour. *Eur. J. Soc. Psychol.* **5**, 1–34 (1975).
58. Jetten, J., Spears, R. & Manstead, A. S. R. Group distinctiveness and intergroup discrimination. in *Social identity: Context, commitment, content* (eds. Ellemers, N., Spears, R. & Doosje, B.) 107–126 (Blackwell Science, 1999).
59. Oakes, P. J. & Turner, J. C. Social categorization and intergroup behaviour: Does minimal intergroup discrimination make social identity more positive? *Eur. J. Soc. Psychol.* **10**, 295–301 (1980).
60. Brown, R. Social Identity Theory: Past achievements, current problems and future challenges. *Eur. J. Soc. Psychol.* **30**, 745–778 (2000).
61. Branscombe, N. R. & Wann, D. L. Collective self-esteem consequences of outgroup derogation when a valued social identity is on trial. *Eur. J. Soc. Psychol.* **24**, 641–657 (1994).

62. Branscombe, N. R., Wann, D. L., Noel, J. G. & Coleman, J. In-Group or Out-Group Extemity: Importance of the Threatened Social Identity. *Pers. Soc. Psychol. Bull.* **19**, 381–388 (1993).
63. Gagnon, A. & Bourhis, R. Y. Discrimination in the Minimal Group Paradigm: Social Identity or Self-Interest? *Pers. Soc. Psychol. Bull.* **22**, 1289–1301 (1996).
64. Rubin, M. & Hewstone, M. Social Identity Theory's Self-Esteem Hypothesis: A Review and Some Suggestions for Clarification. *Personal. Soc. Psychol. Rev.* **2**, 40–62 (1998).
65. Brandt, M. J. & Crawford, J. T. Worldview conflict and prejudice. in *Advances in Experimental Social Psychology* vol. 61 1–66 (Elsevier, 2020).
66. Bergh, R., Akrami, N., Sidanius, J. & Sibley, C. G. Is group membership necessary for understanding generalized prejudice? A re-evaluation of why prejudices are interrelated. *J. Pers. Soc. Psychol.* **111**, 367–395 (2016).
67. Ho, A. K. *et al.* The nature of social dominance orientation: Theorizing and measuring preferences for intergroup inequality using the new SDO₇ scale. *J. Pers. Soc. Psychol.* **109**, 1003–1028 (2015).
68. Esses, V. M., Jackson, L. M. & Armstrong, T. L. Intergroup Competition and Attitudes Toward Immigrants and Immigration: An Instrumental Model of Group Conflict. *J. Soc. Issues* **54**, 699–724 (1998).
69. Riek, B. M., Mania, E. W. & Gaertner, S. L. Intergroup Threat and Outgroup Attitudes: A Meta-Analytic Review. *Personal. Soc. Psychol. Rev.* **10**, 336–353 (2006).
70. Jackson, J. W. Intergroup Attitudes as a Function of Different Dimensions of Group Identification and Perceived Intergroup Conflict. *Self Identity* **1**, 11–33 (2002).

71. Sidanius, J., Pratto, F. & Mitchell, M. In-Group Identification, Social Dominance Orientation, and Differential Intergroup Social Allocation. *J. Soc. Psychol.* **134**, 151–167 (1994).
72. Reynolds, K. J. *et al.* Does personality explain in-group identification and discrimination? Evidence from the minimal group paradigm. *Br. J. Soc. Psychol.* **46**, 517–539 (2007).
73. Perreault, S. & Bourhis, R. Y. Ethnocentrism, Social Identification, and Discrimination. *Pers. Soc. Psychol. Bull.* **25**, 92–103 (1999).
74. Ioannidis, J. P. A. Why Most Published Research Findings Are False. *PLoS Med.* **2**, e124 (2005).
75. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses - Daniël Lakens, 2017. <https://journals.sagepub.com/doi/full/10.1177/1948550617697177>.
76. Goh, J. X., Hall, J. A. & Rosenthal, R. Mini Meta-Analysis of Your Own Studies: Some Arguments on Why and a Primer on How: Mini Meta-Analysis. *Soc. Personal. Psychol. Compass* **10**, 535–549 (2016).
77. Lakens, D. & Etz, A. J. Too True to be Bad: When Sets of Studies With Significant and Nonsignificant Findings Are Probably True. *Soc. Psychol. Personal. Sci.* **8**, 875–881 (2017).
78. Braver, S. L., Thoemmes, F. J. & Rosenthal, R. Continuously Cumulating Meta-Analysis and Replicability. *Perspect. Psychol. Sci.* **9**, 333–342 (2014).
79. Postmes, T., Haslam, S. A. & Jans, L. A single-item measure of social identification: Reliability, validity, and utility. *Br. J. Soc. Psychol.* **52**, 597–617 (2013).

80. Brandt, M. J., Evans, A. M. & Crawford, J. T. The Unthinking or Confident Extremist? Political Extremists Are More Likely Than Moderates to Reject Experimenter-Generated Anchors. *Psychol. Sci.* **26**, 189–202 (2015).
81. Bourhis, R. Y. & Gagnon, A. Social Orientations in the Minimal Group Paradigm. in *Blackwell Handbook of Social Psychology: Intergroup Processes* (eds. Brown, R. & Gaertner, S. L.) 89–111 (Blackwell Publishers Ltd, 2003).
doi:10.1002/9780470693421.ch5.
82. Bourhis, R. Y., Sachdev, I. & Gagnon, A. Intergroup research with the Tajfel matrices: Methodological notes. in *The psychology of prejudice: The Ontario symposium* (eds. Zanna, M. P. & Olson, J. M.) vol. 7 209–232 (Erlbaum, 1994).
83. Brown, R., Tajfel, H. & Turner, J. Minimal group situations and intergroup discrimination: Comments on the paper by Aschenbrenner and schaefer. *Eur. J. Soc. Psychol.* **10**, 399–414 (1980).
84. Turner, J. C. Some comments on... ‘the measurement of social orientations in the minimal group paradigm’. *Eur. J. Soc. Psychol.* **13**, 351–367 (1983).
85. Brewer, M. B., Manzi, J. M. & Shaw, J. S. In-Group Identification as a Function of Depersonalization, Distinctiveness, and Status. *Psychol. Sci.* **4**, 88–92 (1993).
86. Everett, J. A. C. The 12 Item Social and Economic Conservatism Scale (SECS). *PLoS ONE* **8**, e82131 (2013).
87. Robinson, J., Rosenzweig, C., Moss, A. J. & Litman, L. Tapped out or barely tapped? Recommendations for how to harness the vast and largely unused potential of the Mechanical Turk participant pool. *PLOS ONE* **14**, e0226394 (2019).

88. Montgomery, J. M., Nyhan, B. & Torres, M. How conditioning on posttreatment variables can ruin your experiment and what to do about it. *Am. J. Polit. Sci.* **62**, 760–775 (2018).
89. Dunning, D., Meyerowitz, J. A. & Holzberg, A. D. Ambiguity and self-evaluation: The role of idiosyncratic trait definitions in self-serving assessments of ability. *J. Pers. Soc. Psychol.* **57**, 1082–1090 (1989).
90. Kunda, Z. The case for motivated reasoning. *Psychol. Bull.* **108**, 480–498 (1990).
91. Haidt, J. & Graham, J. When Morality Opposes Justice: Conservatives Have Moral Intuitions that Liberals may not Recognize. *Soc. Justice Res.* **20**, 98–116 (2007).
92. Jost, J. T., Sterling, J. & Stern, C. Getting closure on conservatism, or the politics of epistemic and existential motivation. in *The Motivation-Cognition Interface* 56–87 (Taylor and Francis, 2017). doi:10.4324/9781315171388_4.
93. Grieve, P. G. & Hogg, M. A. Subjective Uncertainty and Intergroup Discrimination in the Minimal Group Situation. *Pers. Soc. Psychol. Bull.* **25**, 926–940 (1999).
94. Mullin, B.-A. & Hogg, M. A. Dimensions of subjective uncertainty in social identification and minimal intergroup discrimination. *Br. J. Soc. Psychol.* **37**, 345–365 (1998).
95. Hertel, G. & Kerr, N. L. Priming In-Group Favoritism: The Impact of Normative Scripts in the Minimal Group Paradigm. *J. Exp. Soc. Psychol.* **37**, 316–324 (2001).
96. Duckitt, J., Wagner, C., du Plessis, I. & Birum, I. The psychological bases of ideology and prejudice: Testing a dual process model. *J. Pers. Soc. Psychol.* **83**, 75–93 (2002).
97. Graham, J. *et al.* Mapping the moral domain. *J. Pers. Soc. Psychol.* **101**, 366–385 (2011).
98. Roets, A. & Van Hiel, A. Item selection and validation of a brief, 15-item version of the Need for Closure Scale. *Personal. Individ. Differ.* **50**, 90–94 (2011).

99. Paluck, E. L. & Shepherd, H. The salience of social referents: A field experiment on collective norms and harassment behavior in a school social network. *J. Pers. Soc. Psychol.* **103**, 899–915 (2012).
100. Amiot, C. E. & Bourhis, R. Y. Ideological beliefs as determinants of discrimination in positive and negative outcome distributions. *Eur. J. Soc. Psychol.* **35**, 581–598 (2005).
101. Peters, M., Chisholm, P. & Laeng, B. Spatial Ability, Student Gender, and Academic Performance. *J. Eng. Educ.* **84**, 69–73 (1995).
102. Shepard, R. N. & Metzler, J. Mental Rotation of Three-Dimensional Objects. *Science* **171**, 701–703 (1971).
103. Dovidio, J. F., Kawakami, K. & Gaertner, S. L. Implicit and explicit prejudice and interracial interaction. *J. Pers. Soc. Psychol.* **82**, 62–68 (2002).
104. Kurdi, B., Gershman, S. J. & Banaji, M. R. Model-free and model-based learning processes in the updating of explicit and implicit evaluations. *Proc. Natl. Acad. Sci.* **116**, 6035–6044 (2019).
105. Amodio, D. M. & Ratner, K. G. A Memory Systems Model of Implicit Social Cognition. *Curr. Dir. Psychol. Sci.* **20**, 143–148 (2011).
106. Nosek, B. A. Moderators of the Relationship Between Implicit and Explicit Evaluation. *J. Exp. Psychol. Gen.* **134**, 565–584 (2005).
107. Pinter, B. & Greenwald, A. G. A comparison of minimal group induction procedures. *Group Process. Intergroup Relat.* **14**, 81–98 (2011).
108. Greenwald, A. G., Nosek, B. A. & Banaji, M. R. Understanding and using the Implicit Association Test: I. An improved scoring algorithm. *J. Pers. Soc. Psychol.* **85**, 197–216 (2003).

109. Lakens, D., Scheel, A. M. & Isager, P. M. Equivalence Testing for Psychological Research: A Tutorial. *Adv. Methods Pract. Psychol. Sci.* **1**, 259–269 (2018).
110. Maxwell, S. E., Lau, M. Y. & Howard, G. S. Is psychology suffering from a replication crisis? What does “failure to replicate” really mean? *Am. Psychol.* **70**, 487–498 (2015).
111. Faul, F., Erdfelder, E., Buchner, A. & Lang, A.-G. Statistical power analyses using G*Power 3.1: Tests for correlation and regression analyses. *Behav. Res. Methods* **41**, 1149–1160 (2009).
112. Turner, John C. Social categorization and social discrimination in the minimal group paradigm. in *Differentiation between social groups: Studies in the social psychology of intergroup relations* (ed. Tajfel, H.) 101–140 (Academic Press, 1978).
113. Brown, R. & Bourhis, R. Y. *Instructions for scoring intergroup matrices*. (University of Bristol Mimeo, 1978).
114. Lee, M. D. & Wagenmakers, E.-J. *Bayesian cognitive modeling: A practical course*. (Cambridge University Press, 2013). doi:10.1017/CBO9781139087759.
115. Andrew F. Hayes. PROCESS: A versatile computational tool for observed variable mediation, moderation, and conditional process modeling [White paper]. (2012).
116. Edwards, J. R. & Lambert, L. S. Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychol. Methods* **12**, 1–22 (2007).
117. Preacher, K. J., Rucker, D. D. & Hayes, A. F. Addressing Moderated Mediation Hypotheses: Theory, Methods, and Prescriptions. *Multivar. Behav. Res.* **42**, 185–227 (2007).

118. Schoemann, A. M., Boulton, A. J. & Short, S. D. Determining Power and Sample Size for Simple and Complex Mediation Models. *Soc. Psychol. Personal. Sci.* **8**, 379–386 (2017).
119. Konstantopoulos, S. Fixed effects and variance components estimation in three-level meta-analysis: Three-level meta-analysis. *Res. Synth. Methods* **2**, 61–76 (2011).

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Author Contributions

B.C.R., R.A.A., and A.R.K. designed the pilot and proposed studies. With input from R.A.A. and A.R.K., B.C.R. collected and analyzed pilot data and will collect and analyze the proposed data. B.C.R. wrote the original registered report draft and will write the final paper. R.A.A. and A.R.K. reviewed and edited the registered report and gave final submission approval and will do the same for the final paper.

Competing Interests

The authors declare that they have no competing interests.

Table 1. Design Table

Question	Hypothesis	Sampling plan (e.g. power analysis)	Analysis Plan	Interpretation given to different outcomes
Main Hypothesis Testing				
Is ideology related to <i>explicit</i> minimal group bias? (all studies)	H0: Composite ideology is unrelated to composite intergroup bias H1: Composite ideology is positively related to composite intergroup bias	S1A, S1B, S4: The target sample of 1073 will provide 95% power to detect an effect of $r = .10$ (one-tailed test) S2, S3: The target sample of 1,618 will provide 99.2% power to detect an effect of $r = .10$ (one-tailed test)	We will examine the correlation between composite ideology and composite intergroup bias	H0 will be considered supported if the association between ideology and intergroup bias is $p > .05$ or $r < .10$ and an inferiority test using the 90% CI around the observed correlation does not include or exceed .1.

				H1 will be considered supported if the correlation is positive and statistically significant at $p < .05$ and the size of the association is $r = .10$ or above.
Is ideology related to <i>implicit</i> minimal group bias? (Study 4)	H0: Composite ideology is unrelated to implicit intergroup bias H1: Composite ideology is positively related to implicit intergroup bias	The target sample of 1073 will provide 95% power to detect an effect of $r = .10$ (one-tailed test)	We will examine the correlation between composite ideology and implicit intergroup bias (IAT scores)	H0 will be considered supported if the association between ideology and implicit intergroup bias is $p > .05$ or $r < .10$ and an inferiority test using the 90% CI around the observed correlation does not include or exceed .1. H1 will be considered supported if the correlation is positive and statistically significant at $p < .05$ and the size of the association is $r = .10$ or above.
Moderation Analyses				
Does sample moderate ideological differences in minimal group bias? (combined S1A and S1B datasets)	H0: Sample does not moderate the relationship between ideology and minimal group bias H1: Sample does moderate the relationship between ideology and minimal group bias	The target sample of 2146 will provide more than 95% power to detect $\beta = .10$.	We will regress a sample type categorical moderator variable and continuous composite ideology score and their interaction onto composite intergroup bias	H0 will be considered supported if the interaction is $p > .05$ or the difference in slopes is $\beta < .10$ and an inferiority test using the 90% CI around the observed difference in slopes does not include or exceed .1 if positive or -.1 if negative.

				<i>Significant (either positive or negative):</i> H1 will be considered supported if the interaction is significant at $p < .05$ and the size of the difference in slopes (ideology-bias) between samples is $\beta = .10$ or above if positive or $\beta = -.10$ or below if negative.
Does group type (random versus volitional versus attitude-similarity) moderate ideological differences in minimal group bias? (S2)	H0: Group type does not moderate the relationship between ideology and minimal group bias H1: Group type does moderate the relationship between ideology and minimal group bias	The target sample of 4854 will provide 95% power to detect $\beta = .10$.	We will regress dummy-coded variables for the volitional group and the attitude-similarity group and the continuous composite ideology score and their interactions onto composite intergroup bias	H0 will be considered supported if the interactions are $p > .05$ or the differences in slopes is $\beta < .10$ if positive or $\beta > -.10$ if negative and an inferiority test using the 90% CI around the observed difference in slopes does not include or exceed .1 if positive or -.1 if negative. H1 will be considered supported if at least one of the interactions is significant at $p < .05$ and the size of the difference in slopes (ideology-bias) between conditions is $\beta = .10$ or above if positive or $\beta = -.10$ or below if negative.

Does group type (control versus cooperative versus competitive) moderate ideological differences in minimal group bias? (S3)	H0: Group type does not moderate the relationship between ideology and minimal group bias H1: Group type does moderate the relationship between ideology and minimal group bias	The target sample of 4854 will provide 95% power to detect $\beta = .10$.	We will regress dummy-coded variables for the cooperative group and the competitive group and the continuous composite ideology score and their interactions onto composite intergroup bias	H0 will be considered supported if the interaction is $p > .05$ or the difference in slopes is $\beta < .10$ if positive or $\beta > -.10$ if negative and an inferiority test using the 90% CI around the observed difference in slopes does not include or exceed .1 if positive or -.1 if negative. H1 will be considered supported if at least one of the interactions is significant at $p < .05$ and the size of the difference in slopes (ideology-bias) between conditions is $\beta = .10$ or above if positive or $\beta = -.10$ or below if negative.
Mediation Analyses				
What mediates the relation between ideology and intergroup bias? (S2)	H0: none of our mediator variables significantly mediates this relation H1: one or more of our mediator variables significantly mediates this relation.	The target sample of 4854 will provide >95% power to detect mediation of an effect of size $r = .10$ between ideology and intergroup bias, given literature on the expected correlations between ideology and our proposed mediators ($r_s > .30$) and between our mediators and intergroup bias ($r_s > .20$)	We will conduct a mediation analysis using the PROCESS macro for SPSS, with composite ideology as the IV, composite intergroup bias as the DV, and each of the three proposed mediators (ingroup-protective morality, intolerance of uncertainty, and perceived norms) simultaneously	H0 will be considered supported if all of the 95% confidence intervals of the indirect effects contain 0. In light of existing theory, we cannot interpret a negative mediation effect – i.e., if one or more of the 95% confidence intervals of the indirect effects do

			entered as mediator variables.	not contain 0 and are negative. H1 will be considered supported if one or more of the 95% confidence intervals of the indirect effects do not contain 0 and are positive.
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Currently in revision, please do not circulate